In-Home Assistive Robot

Week 9 Schema and Technology Stack | November 11, 2026

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This document consolidates D1's call schema and stack with D2's interface refinements and the Senior Design prototype evidence. It is a current reconstruction, not a claim that a Week 9 submission or production database already existed. Austin owns software/data; Gavin checks resident/caregiver meaning and accessibility; Liam checks sensing/platform records. Their component reviews remain to be recorded.

Schema and purpose

The D1 relational model has four entities. Robot and Caregiver have a many-to-many relationship through Pairing; one Pairing owns zero or more CallAttempts. Each attempt belongs to exactly one robot/caregiver pair. Pairing is call eligibility, while current resident CALL/FOLLOW consent remains local and volatile. D1 section 2 and entity diagram.

Entity / field	Type	Key, nullability and rule	Purpose
Robot.robot_id	Text	PK; 1-64 allowed ASCII ID characters	Enrolled robot identity
Robot.last_seen_utc	UTC text or null	Latest accepted heartbeat; RFC 3339	Display online age, not location
Robot.last_sequence	Integer or null	0..2^63-1; increasing within authenticated session	Reject repeated/out-of-order heartbeats
Caregiver.caregiver_id	Text	PK; identity-provider subject	Account identity without local password
Caregiver.display_name	Text	1-80 characters	Show named caller before acceptance
Pairing.robot_id	Text	Composite PK; FK to Robot	Associate account with robot
Pairing.caregiver_id	Text	Composite PK; FK to Caregiver	Associate robot with account
Pairing.active	Boolean	True/false; inactive denies new call access	Revoke eligibility without erasing old calls
CallAttempt.call_id	UUID text	PK; canonical UUID	Identify one attempt
CallAttempt.robot_id	Text	Composite FK with caregiver_id to Pairing	Preserve owner robot
CallAttempt.caregiver_id	Text	Composite FK with robot_id to Pairing	Preserve owner account
CallAttempt.request_id	UUID text or null	Unique with caregiver_id when supplied	Deduplicate retried start
CallAttempt.status	Enum text	ringing, connected, declined, no_answer, offline, ended, failed	Truthful communication outcome
CallAttempt.created_utc	UTC text	Required RFC 3339	Attempt creation
CallAttempt.accepted_utc	UTC text or null	Null until resident acceptance	Distinguish acceptance from transport connection
CallAttempt.ended_utc	UTC text or null	Null until closure	Terminal outcome time

Integrity and lifecycle

Index calls on (robot_id, created_utc); keep (caregiver_id, request_id) unique. A partial unique index on robot_id for ringing/connected calls admits only one active attempt per robot. Start/end/outcome transitions are transactional. A repeated request returns the original result; a retry ID reused with different input is a conflict. Decline and other terminal outcomes never become connected after a delayed event. On restart, active attempts close as failed and the robot remains offline until a fresh authenticated session/heartbeat.

Only the call service opens the SQLite file. Audio/video, access tokens, passwords, masks, medical information and resident trajectories do not belong in this schema. The proposed 30-day call history requires a tested purge and backup policy; an inactive pair is pruned only after dependent records are removed. Engineering configuration is a separate versioned local record, and seven-day diagnostic metadata is a separate bounded store. Privacy inventory and deletion limits.

Consent and transient contracts

D2 Context binds boot UUID, call UUID, epoch and sequence. CallLease adds audio/video scopes, local expiry and environment revision; FOLLOW and target selection have a separate epoch. Sensor/frame buffers, masks, SDP/ICE and media remain bounded volatile data. No stored pairing, call record or model result restores permission after reboot/revoke. D1's optical-depth TrackRecord remains offline with motion_authorized=false; D2 RobotTargetV2 requires a qualified producer in calibrated robot coordinates, which has not been established. D2 section 2.

Technology stack

The table distinguishes **observed** local source, **reported** Space artifacts, **proposed** D1/D2 integrations and **research alternatives**. These categories prevent a prototype runtime version from becoming an invented production version. A version marked unpinned must be fixed in manifests/lockfiles before integration or release.

Layer	Technology / version evidence	Status and owner	Reason and boundary
Local call bench	Node; package 0.1.0; Node 23.7.0 in its README, engines >=22.5	Observed local source; Austin	Native HTTP/SSE, crypto and optional node:sqlite; no npm dependencies declared. Loopback only.
Browser media/UI	Native WebRTC, HTML/CSS/JavaScript; exact Chrome build unrecorded here	Observed source; Austin/Gavin	Generated test media on one Mac; physical camera/mic, LAN and remote deployment unqualified.
One-shot reminders	Node 24.22.1 tested; native SQLite worker; extended package revision in Space	Reported separate artifact; Austin/Gavin	Exact-version resident approval, one future occurrence, explicit sharing and truthful self-report. Not a replacement for D1 call schema.
Robot application	Python, version unpinned	Proposed D1/D2; Austin	Typed local adapters and consent/control contracts; keep blocking work outside control deadlines.
Call API/signaling	FastAPI, Uvicorn, Pydantic; unpinned	Proposed D1; Austin	HTTP/WSS, strict validation and transactional outcome state machine. Not the current Node bench.
Database	SQLite, product version unpinned	Proposed call service; Austin	Single-server relational constraints; compare PostgreSQL only if measured scale requires it.
Identity	Keycloak + Authlib, unpinned	Proposed; Austin	OIDC/PKCE, issuer/audience/expiry/subject checks and scoped pairing. Fixture role labels are not this integration.
Caregiver companion	Swift/SwiftUI, SDK versions unpinned	Proposed; Austin/Gavin	Native system sign-in, Keychain refresh storage, scalable text/VoiceOver. No complete mobile call validation.
Baseline perception	PyTorch/TorchVision masks; NumPy optional; versions/weights unpinned	Offline experiments; Austin	Code license differs from weight/data rights. CPU timing does not establish onboard real-time performance.
SAM tracking direction	SAM3 family; direct Meta sam-3.1 route documented in research; exact selected provider/version unpinned	Research/integration candidate; Austin	Selected masks are not identity, depth or movement authority. Upload retention and result freshness need route-specific checks.
Simulation/reference	MuJoCo; Open Beni source/model/checkpoint references	Separate research artifacts; Austin/Liam	Reproduction and bounded experiments only; asset licenses and hashes must follow the exact artifact. No hardware release claim.
Motion platform	Dedicated local controller over bounded USB serial; board/firmware unselected	Proposed D2; Liam/Austin	Local watchdog, sequence/boot/expiry checks, feedback and separately qualified caster/balancing profiles.
Application compute	Existing computer for bench; Pi/Jetson/nearby-host comparisons	Alternatives, not purchase decisions	Choose from concurrent workload, encoding, thermal, mass and power measurements.
Hosting/network	Ubuntu VM, HTTPS/WSS, optional relay; provider/build unselected	Proposed; Austin	Authenticated outbound robot signaling; no insecure fallback. Loopback ICE rewriting cannot be reused on a LAN.
Middleware/inference runtimes	ROS 2, ONNX Runtime or MNN	Research alternatives only	Add only for a concrete integration need and measured compatibility. They are not adopted dependencies here.

Current evidence and next integration

The inspected local call package is at /Users/austinheiss/Documents/BeniResearch/stationary_call_20261109. Its README and manifests identify eight generated-media calls, a native-module-only package and an imported fixture authority. The Space integration record separately reports the newer routine extension and its checks; that package was not represented as installed in this repository. Its call/routine authority, presence and durable routine records must not be silently merged into D1's call service model.

Continue with one canonical authority and a narrow robot adapter; preserve behavior across a later framework port. Local Stop/Privacy, capture ownership and deadline checks remain onboard. Offboard inference returns observations/proposals bound to original capture IDs, not raw motor commands. Synchronous optional terminal-call journaling on the Node event loop is a recorded gap; changing frameworks alone does not resolve it. Runtime implementation research.

Cross references and approval

The ethics statement, NOTICE and constraints essay use this same baseline. Austin confirms team ownership of the architecture/model/priorities/rationale. This consolidation selects no new architecture or hardware; owner review, deployed settings and individual signatures remain separate evidence.